

Project work for the teaching *Time Series Analysis*

Master in *Data Science and Statistical Learning* (MD2SL), A.Y. 2023/2024

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The student must retrieve **two distinct time series** from a publicly available source.

- **The first time series must show no seasonality**, for instance it may have annual frequency, have been deseasonalised, or be a financial time series.
- **The second time series must have frequency less than one year and be characterized by a seasonal pattern.**

Ideally, the first time series should consist of at least 50 measurements, and the second time series should span over at least 10 years. Also, the two time series should contain no more than 5% missing values. Outliers with high leverage on the estimates can be set to missing or preliminarily imputed making use of any imputation procedure. **The time series cannot be simulated or already analysed in class, and cannot be taken from software packages unless they also appear in a research paper.** The student must provide a clear reference to the source of the two time series.

For each time series, the student must prepare:

- **a report of at most 10 pages** with font 14 *Times New Roman* addressing the 10 issues below. The use of R for Statistical Computing is encouraged, but the student is free to use any other software. The report may include formulas, tables and figures but no code or reference to the software;
- **a zip file including everything that reproduces the results included in the report** (data, R code, Excel files, etc.).

Assessment

- Each issue is evaluated for a maximum of 3 points, thus the grade on the project work ranges between 0 and 30.

Issues to be addressed for the first time series

1. Describe the data, explaining which phenomenon they represent and providing details on the measurements (time span, frequency, scale). Display the time series and discuss its appearance in reference to stationarity and to the presence of outliers.
2. Find an **ARIMA model** and an **ARMA model with linear deterministic trend** that fit adequately to the time series. To do this, make use of a combination of graphical checks, significance tests and automated search. Some guidelines:

- for the ARIMA model, evaluate whether a drift is necessary;
 - make use of transformations (logarithm, squared root or automated Box-Cox) and/or reduce the window of observation (just in case of evident heteroschedasticity) to improve the adequacy of the model;
 - if deemed necessary, detect and replace outliers manually or using an automated procedure.
3. Provide a formal definition (with formulas) of the two models, emphasizing the different theoretical properties, then report a summary of parameter estimates for each of them.
 4. For each of the two models, report and discuss some opportune diagnostics to check their adequacy.
 5. Estimate the forecast accuracy of each of the two models at some horizons in the future through rolling window cross-validation, then provide a graphical comparison and discuss the differences as a function of the horizon.
 6. Display forecasts of future values of the time series based on each of the two models, then discuss the difference between them in light of their theoretical properties (e.g., presence of deterministic and/or stochastic trend).

Issues to be addressed for the second time series

1. Describe the data, explaining which phenomenon they represent and providing details on the measurements (time span, frequency, scale). Display the time series and discuss its appearance in reference to stationarity and to the presence of seasonal patterns and outliers.
2. Find a **seasonal ARIMA model** that fit adequately to the time series. To do this, make use of a combination of graphical checks, significance tests and automated search. Some guidelines:
 - evaluate whether a drift is necessary;
 - try to apply no more than one non-seasonal and one seasonal difference;
 - make use of transformations (logarithm, squared root or automated Box-Cox) and/or reduce the window of observation (just in case of evident heteroschedasticity) to improve the adequacy of the model;
 - if deemed necessary, detect and replace outliers using an automated procedure.
3. Provide a formal definition (with formulas) and a description of the model, emphasizing its theoretical properties, then report a summary of parameter estimates.

4. Report and discuss some opportune diagnostics to check the adequacy of the model. Estimate and discuss the forecast accuracy of the model at some horizons in the future through rolling window cross-validation.

Suggested data sources

- ISTAT <https://dati.istat.it/>
- EUROSTAT <https://ec.europa.eu/eurostat/data/database>
- OECD <https://data.oecd.org/>
- The World Bank <https://data.worldbank.org/>
- Faostat <https://www.fao.org/faostat/en>
- Yahoo Finance <https://it.finance.yahoo.com>
- Kaggle <https://www.kaggle.com/datasets>

Suggested R functions

- Package ‘tseries’: `adf.test`, `kpss.test`
- Package ‘forecast’: `Acf`, `Pacf`, `Box.test`, `Arima`, `auto.arima`, `BoxCox.lambda`, `forecast`, `tsCV`, `tsoutliers`
- Package ‘TSA’: `BoxCox.ar`